Refactoring.to eclipse collections

Making Your Java Streams Leaner, Meaner, and Cleaner

Donald Raab, Vladimir Zakharov

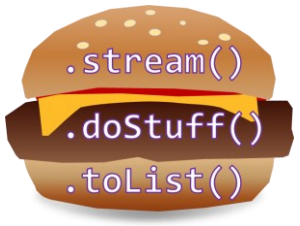
dev2next • Colorado Springs, CO • September 29 - October 2, 2025

Introduction

- What is Eclipse Collections?
  - Feature rich, memory efficient Java Collections framework
- History
  - Eclipse Collections started off as an internal collections framework named Caramel at Goldman Sachs in 2004

  - In 2012, it was open sourced to GitHub as a project called GS Collections
  - GS Collections was migrated to the Eclipse Foundation, re-branded as Eclipse Collections in 2015
  - Eclipse Collections 12.0 (Java 11+) & 13.0 (Java 17+) released in June 2025
- Learn Eclipse Collections with Kata
- Eclipse Collections is open for contributions!

Why Refactor to EC?



Lazy or Eager

TestUtils

Testability/Debuggability

No “Bun” methods

Drop-in replacement for JDK types

Concise Code

Benefits

It’s Easy!

Intuitive humane APIs

Rich[er] APIs

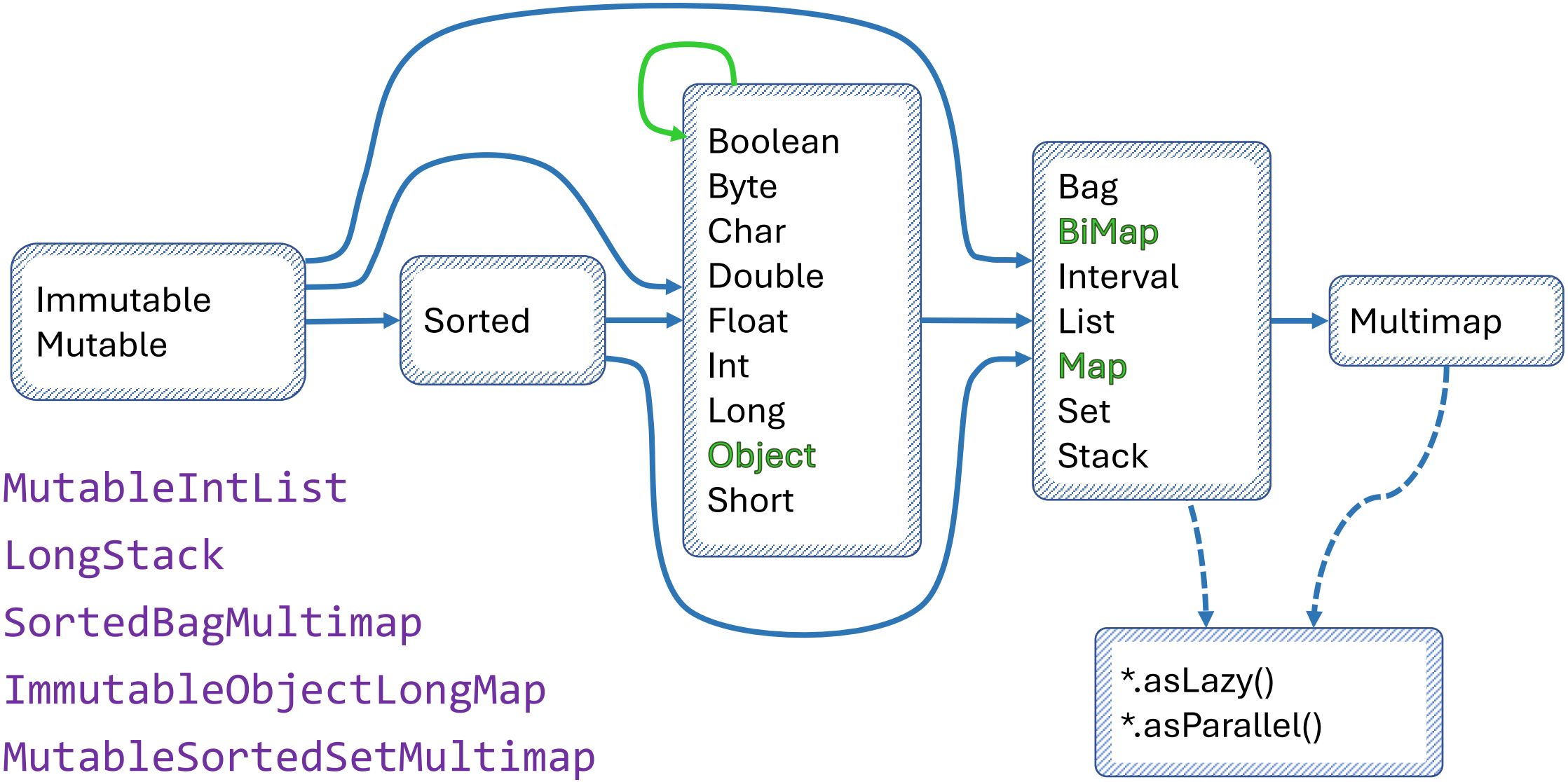
Memory Savings

Better Performance [*citation needed*]

Comprehensive Support for Primitive Types

Efficient Maps and Sets

Any Types You Need



`MutableIntList`

`LongStack`

`SortedBagMultimap`

`ImmutableObjectLongMap`

`MutableSortedSetMultimap`

Instantiate Them Using Factories

Primitive Type*	Container Type	Mutability	Multimap	Initialized	Lazy
Boolean	Bags	.mutable	.bag	.empty()|.of()|.with()	.asLazy()
Byte	BiMaps	.immutable	.list	.of(one)|.with(one)	
Char	Lists	.fixedSize	.set	...	
Double	Maps		.sortedSet	.of(one,…,ten)	
Float	Multimaps			.with(one,…,ten)	
Int	Sets			.of(... elements)	
Long	SortedBags			.with(... elements)	
Object	SortedMaps			.ofAll(Iterable)	
Short	SortedSets			.withAll(Iterable)	
	Stacks				

* *Pick two for maps*

```
ImmutableLongStack
LongStacks.immutable.with(1, 2, 3).asLazy()
LazyLongIterable
```

Evolving from JDK to EC Factories and Types

```
// JDK Type
Set<String>
```

Mutable

Immutable

How do you know if a collection is mutable or immutable?

```
// JDK Type
Set<String>
```

```
// JDK Implementation
new HashSet<>(List.of("1", "2", "3"))
```

```
// JDK Type
Set<String>
```

```
// EC Implementation
Sets.mutable.of("1", "2", "3")
```

```
// EC Type
MutableSet<String>
```

```
// EC Implementation
Sets.mutable.of("1", "2", "3")
```

Start

Step 1

Step 2

```
// JDK Type
Set<String>
```

```
// JDK Implementation
Set.of("1", "2", "3")
```

```
// JDK Type
Set<String>
```

```
// EC Implementation
Sets.immutable.of("1", "2", "3")
    .castToSet()
```

```
// EC Type
ImmutableSet<String>
```

```
// EC Implementation
Sets.immutable.of("1", "2", "3")
```

Evolving from JDK to EC Factories and Types

```
// JDK Type
List<String>
```

Mutable

Immutable

How do you know if a collection is mutable or immutable?

```
// JDK Type
List<String>
```

```
// JDK Implementation
new ArrayList<>(List.of("1", "2", "3"))
```

```
// JDK Type
List<String>
```

```
// JDK Implementation
List.of("1", "2", "3")
```

Start

```
// JDK Type
List<String>
```

```
// EC Implementation
Lists.mutable.of("1", "2", "3")
```

```
// JDK Type
List<String>
```

```
// EC Implementation
Lists.immutable.of("1", "2", "3")
    .castToList()
```

Step 1

```
// EC Type
MutableList<String>
```

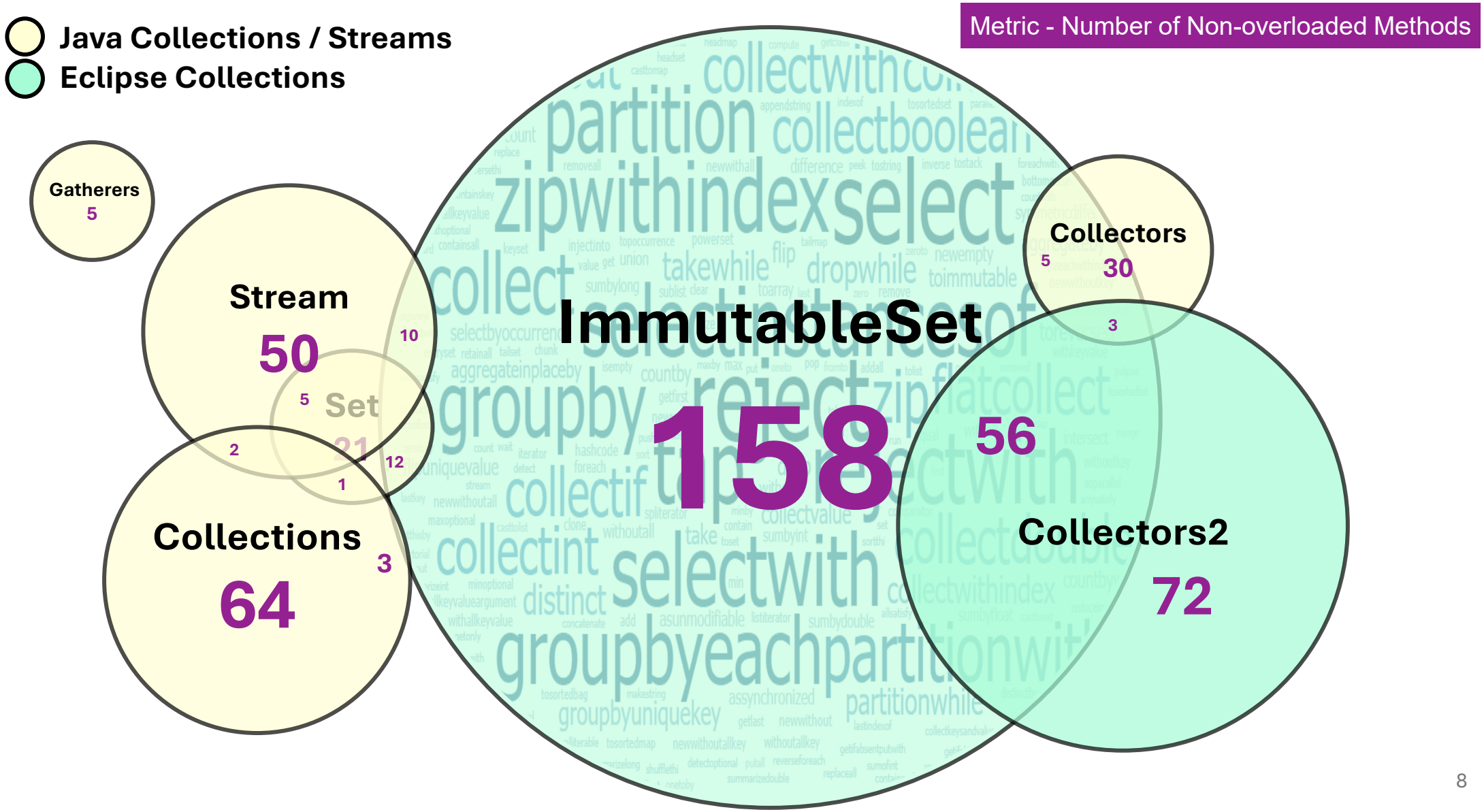
```
// EC Implementation
Lists.mutable.of("1", "2", "3")
```

```
// EC Type
ImmutableList<String>
```

```
// EC Implementation
Lists.immutable.of("1", "2", "3")
```

Step 2

Set/Stream vs. ImmutableSet



Methods by Category - Highlights

transforming

collect[With]
collect[Boolean,Byte,Char,Double,Float,Int,Long,Short]
collectIf
collectKeysAndValues
collectValues
collectWithIndex
collectWithOccurrences
flatCollect

grouping

groupBy
groupByEach
groupByUniqueKey
sumBy[Double,Float,Int,Long]
sumOf[Double,Float,Int,Long]
aggregateBy
aggregateInPlaceBy

converting

toArray
toBag
toImmutable
toList
toMap
toReversed
toSet
toSortedBag[By]
toSortedList[By]
toSortedMap
toSortedSet
toSortedSet[By]
toStack
toString

filtering

select[With]
selectByOccurrences
selectInstancesOf
reject[With]
partition[With]
partitionWhile

wrapping

asLazy
asParallel
asReversed
asSynchronized
asUnmodifiable

finding

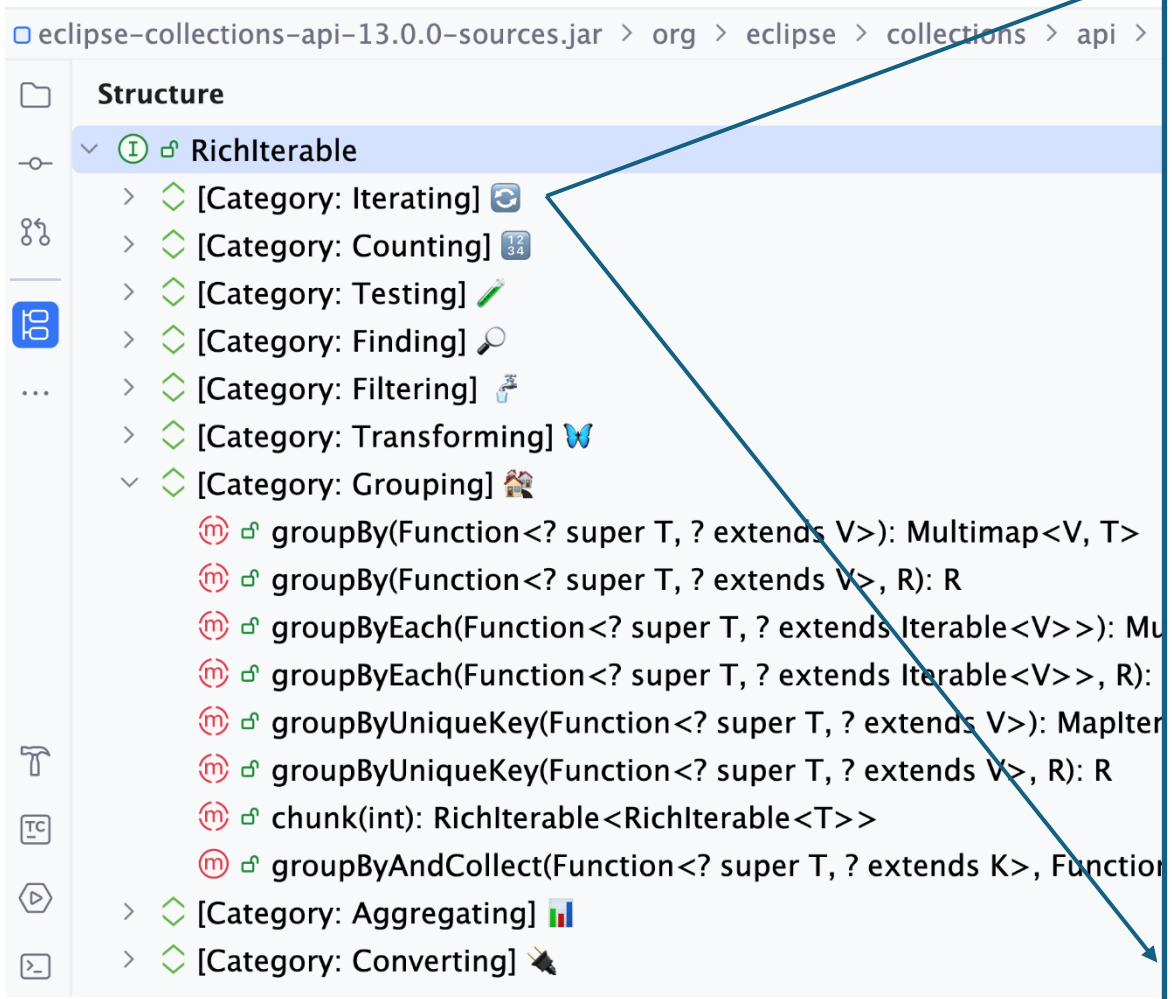
detect[With]
detect[With]IfNone
detect[With]Optional
max[By]
min[By]

testing

allSatisfy[With]
anySatisfy[With]
noneSatisfy[With]
notEmpty
isEmpty

Simulating Method Categories in IntelliJ

- Using **Custom Code Folding Regions**




```
//region [Category: Iterating]

@Override
default void forEach(Procedure<? super

/** The procedure is executed for each
/UnnecessaryFullyQualifiedName/
void each(Procedure<? super T> procedu

/** Executes the Procedure for each e
RichIterable<T> tap(Procedure<? super

/** Returns a lazy (deferred) iterable
LazyIterable<T> asLazy();

//endregion [Category: Iterating]
```

Let’s Do It!

Memory: Methodology

Java Object Layout (JOL) - OpenJDK Code Tools Project

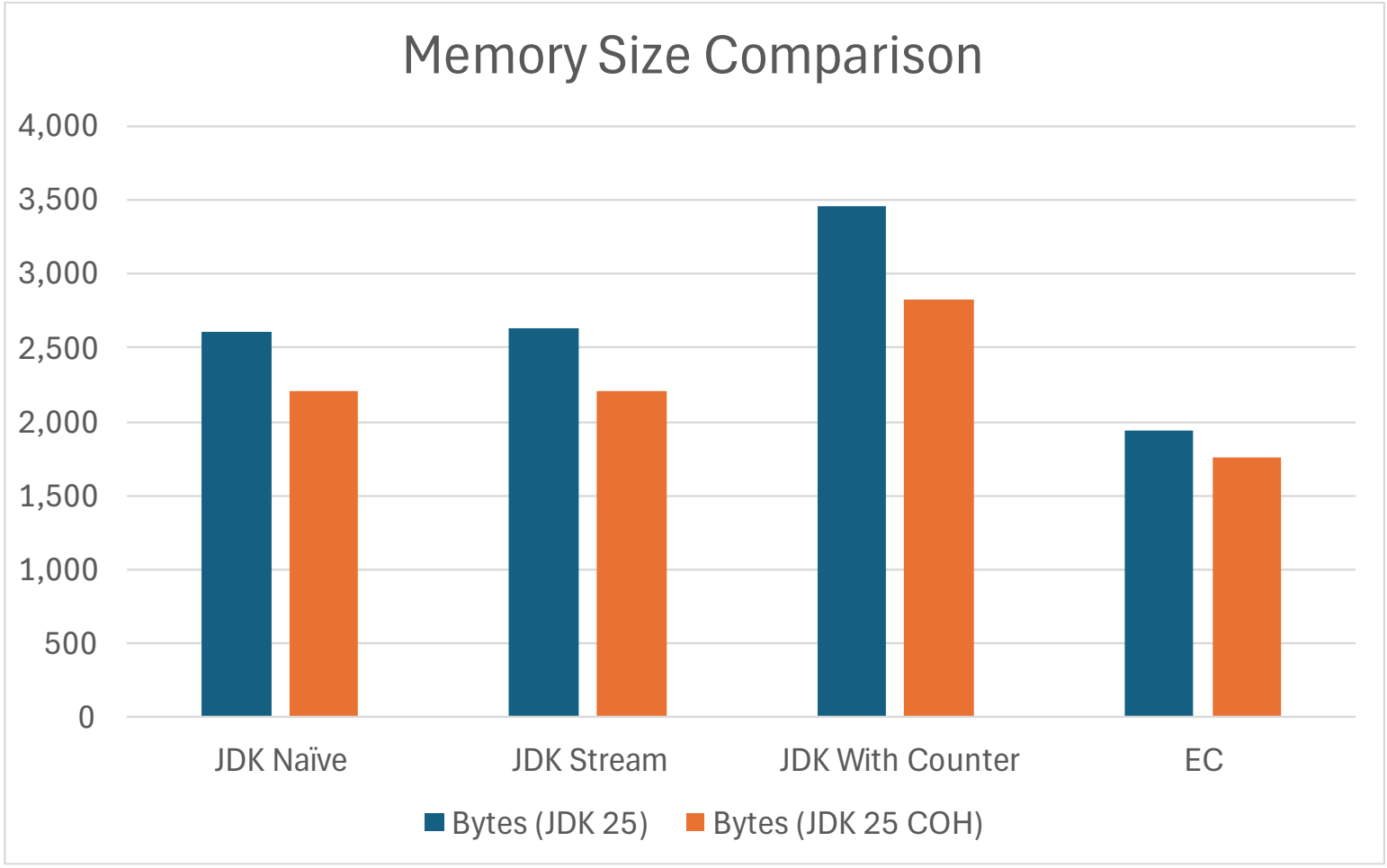
JOL (Java Object Layout) is the tiny toolbox to analyze object layout schemes in JVMs. JOL [is] much more accurate than other tools relying on heap dumps, specification assumptions, etc.

```
<dependency>
    <groupId>org.openjdk.jol</groupId>
    <artifactId>jol-core</artifactId>
    <version>0.17</version>
</dependency>
```

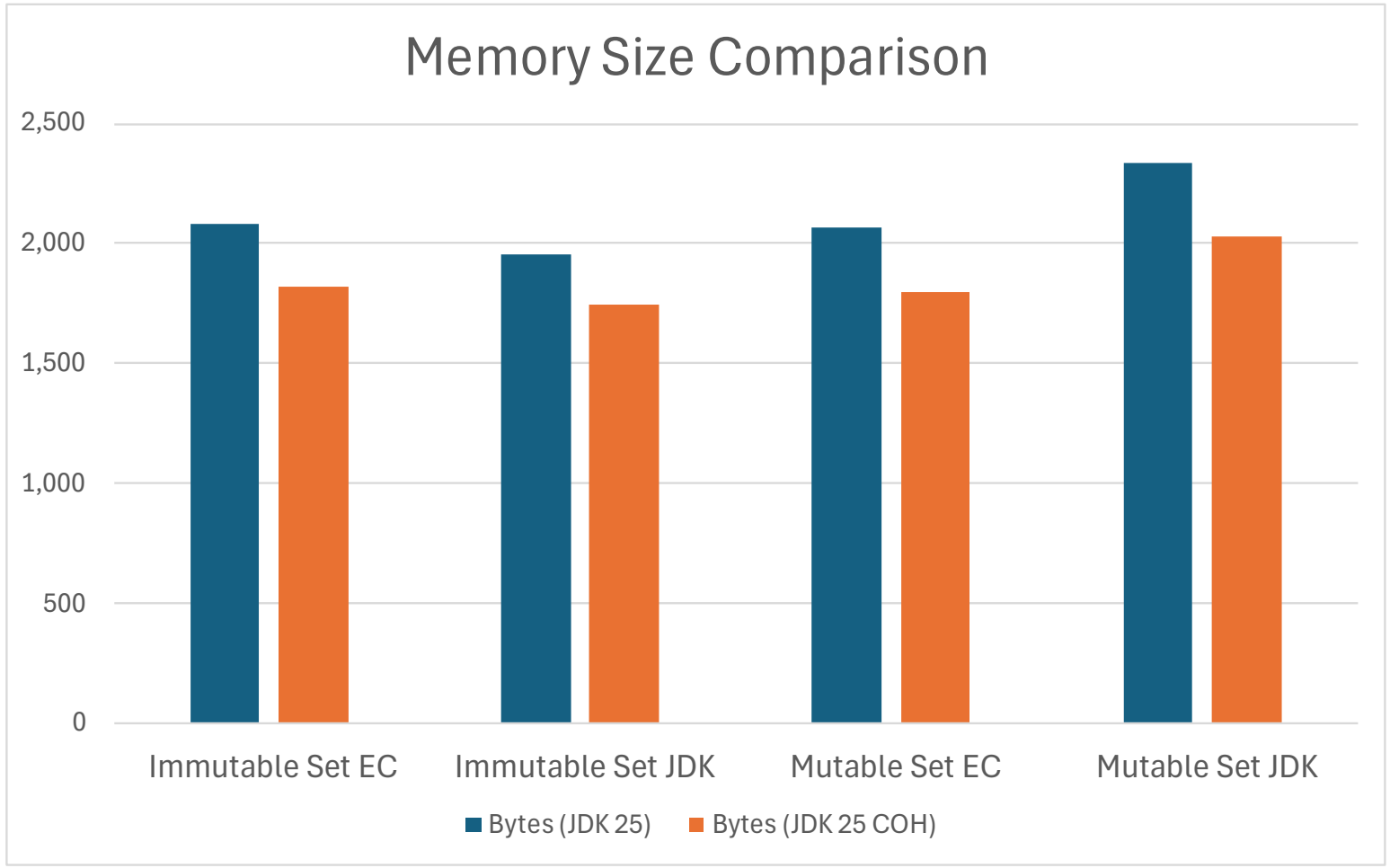
Further reading

- Baeldung: Memory Layout of Objects in Java
- StackOverflow:
  - Does Java Object Layout work with Java Records?
  - Does Java Object Layout (JOL) work with Java 25 and Compact Object Headers enabled?

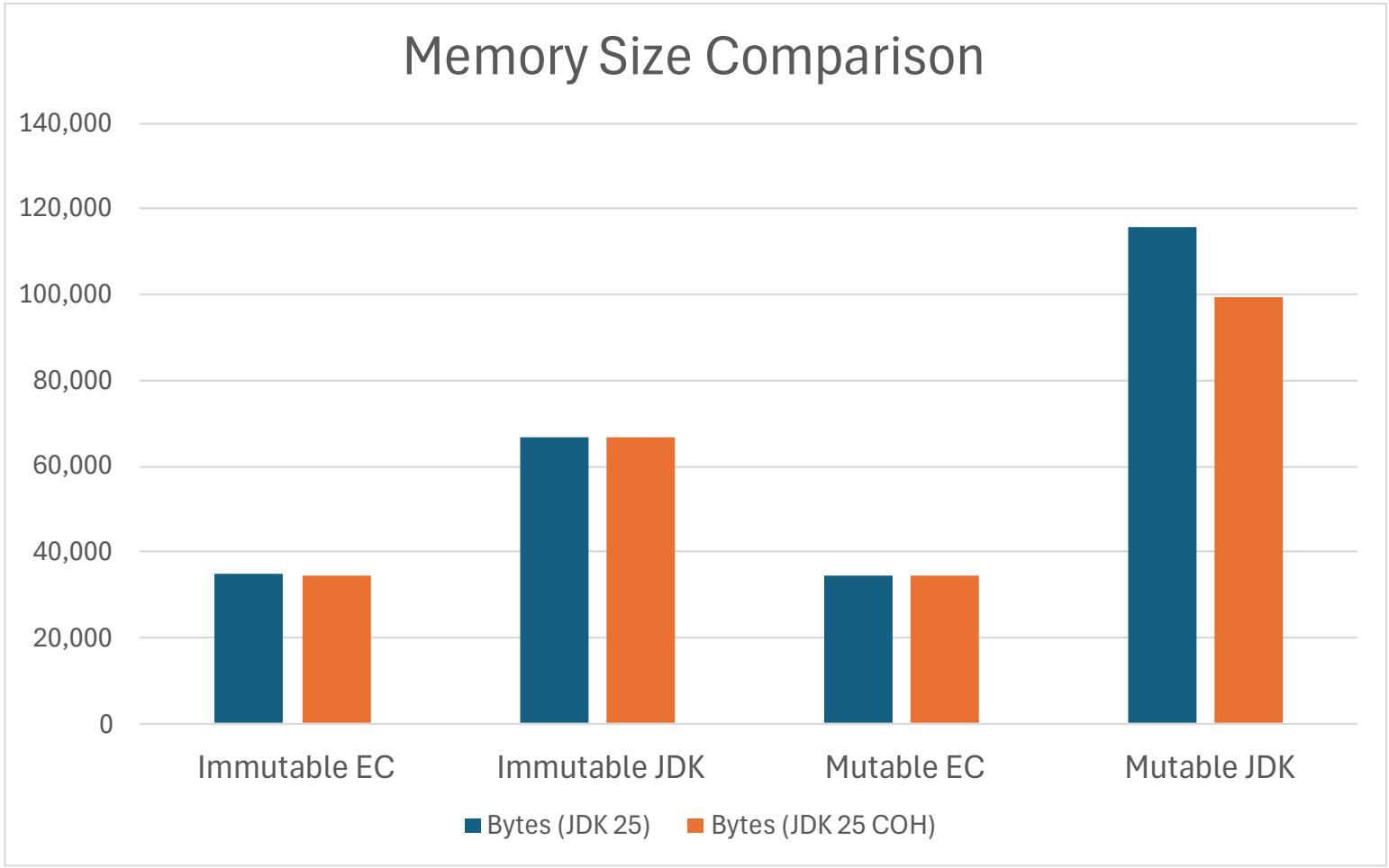
Memory: Word → Count Map



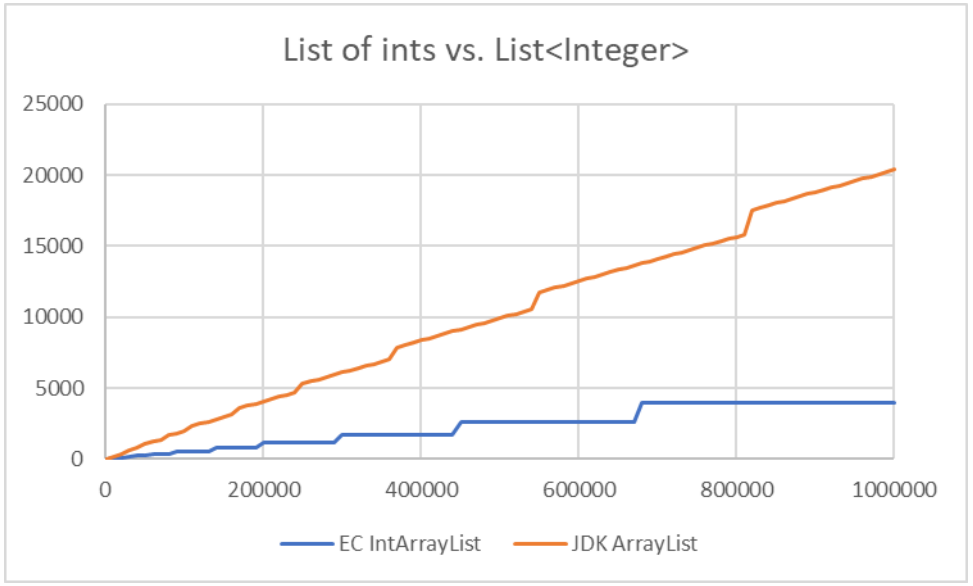
Memory: Generation Set

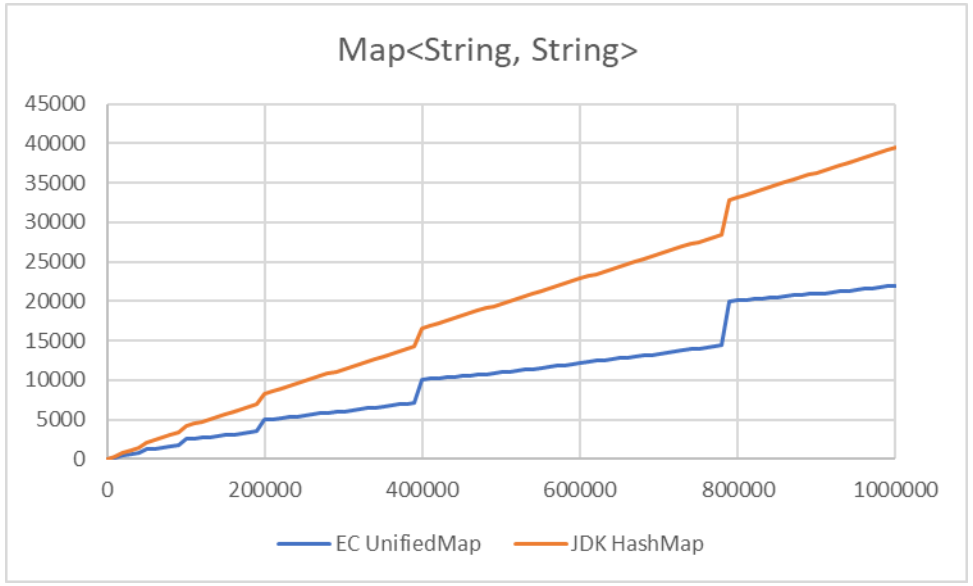


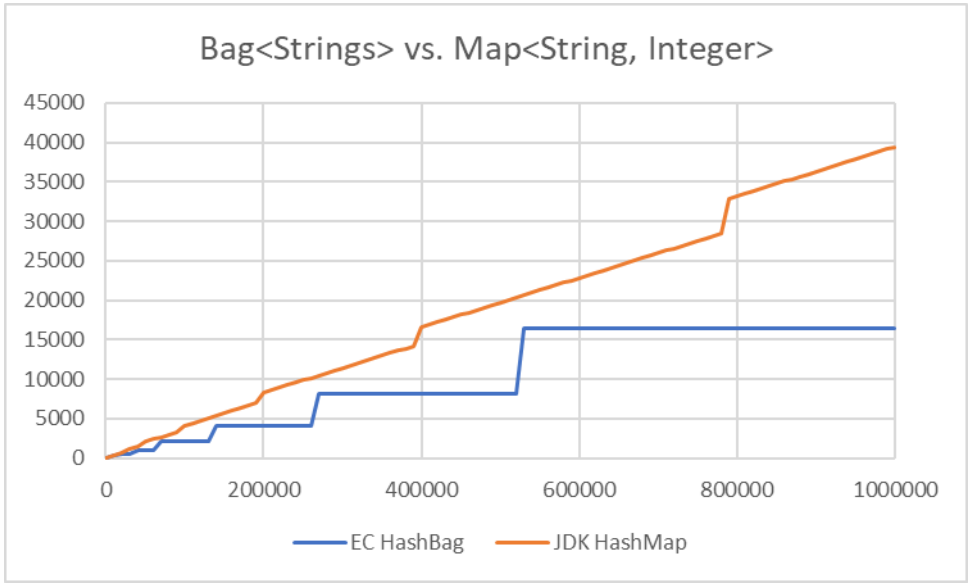
Memory: Year → Generation Map

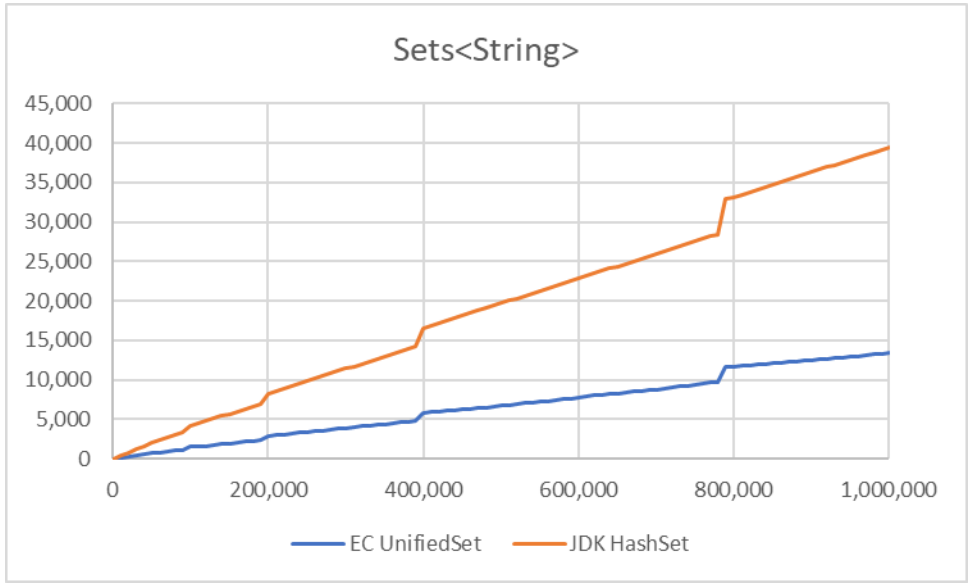


Memory Usage (Overhead in KB, Count)









Eclipse Collections – What's in It for Me?

Features	Productivity	Performance	Thread Safety	Type Safety
Rich functional API with good symmetry	✓			
Memory Efficiency		✓		
Optimized Eager API		✓		
Primitive Collections	✓	✓		✓
Immutable Collections		✓	✓	
Lazy and Parallel Lazy API		✓		
Multimap, Bag, BiMap, Pool, Interval		✓		✓
Hashing Strategy Data Structures	✓	✓		
Multi-reader Data Structures		✓	✓	
Mutable and Immutable Collection Factories	✓			

Example: DataFrame-EC

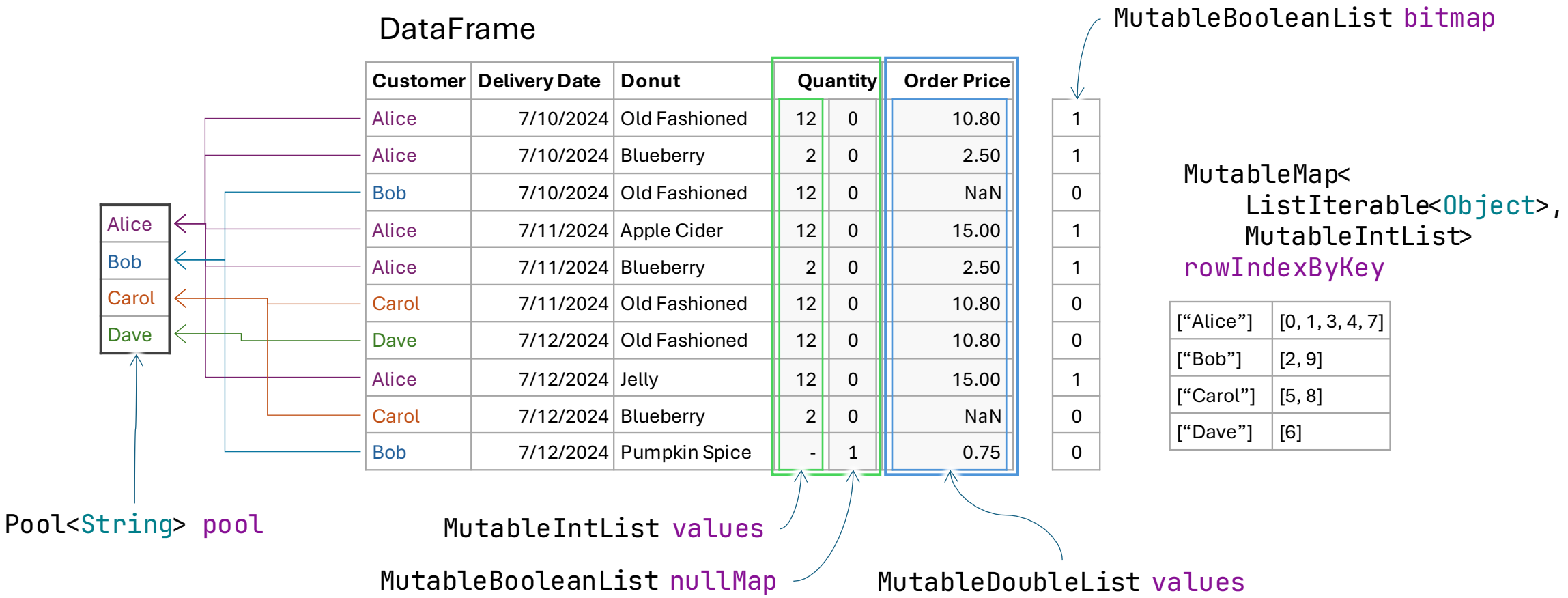
…built on the solid foundation of Eclipse Collections types

DataFrame

Customer	Delivery Date	Donut	Quantity	Order Price
Alice	7/10/2024	Old Fashioned	12	10.80
Alice	7/10/2024	Blueberry	2	2.50
Bob	7/10/2024	Old Fashioned	12	null
Alice	7/11/2024	Apple Cider	12	15.00
Alice	7/11/2024	Blueberry	2	2.50
Carol	7/11/2024	Old Fashioned	12	10.80
Dave	7/12/2024	Old Fashioned	12	10.80
Alice	7/12/2024	Jelly	12	15.00
Carol	7/12/2024	Blueberry	2	null
Bob	7/12/2024	Pumpkin Spice	null	0.75

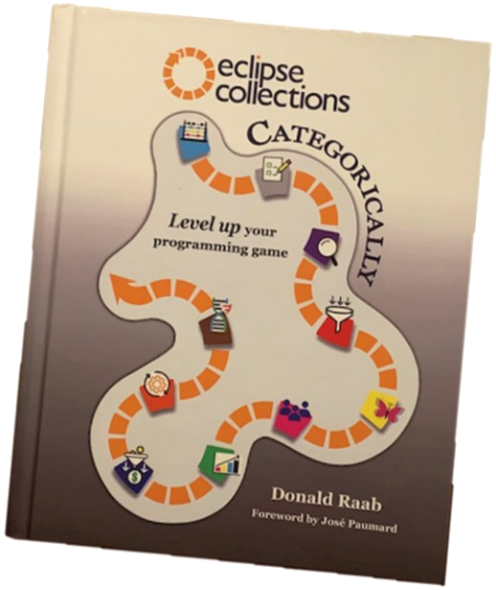
Example: DataFrame-EC

...the foundation, and walls, and joints, and studs, and...



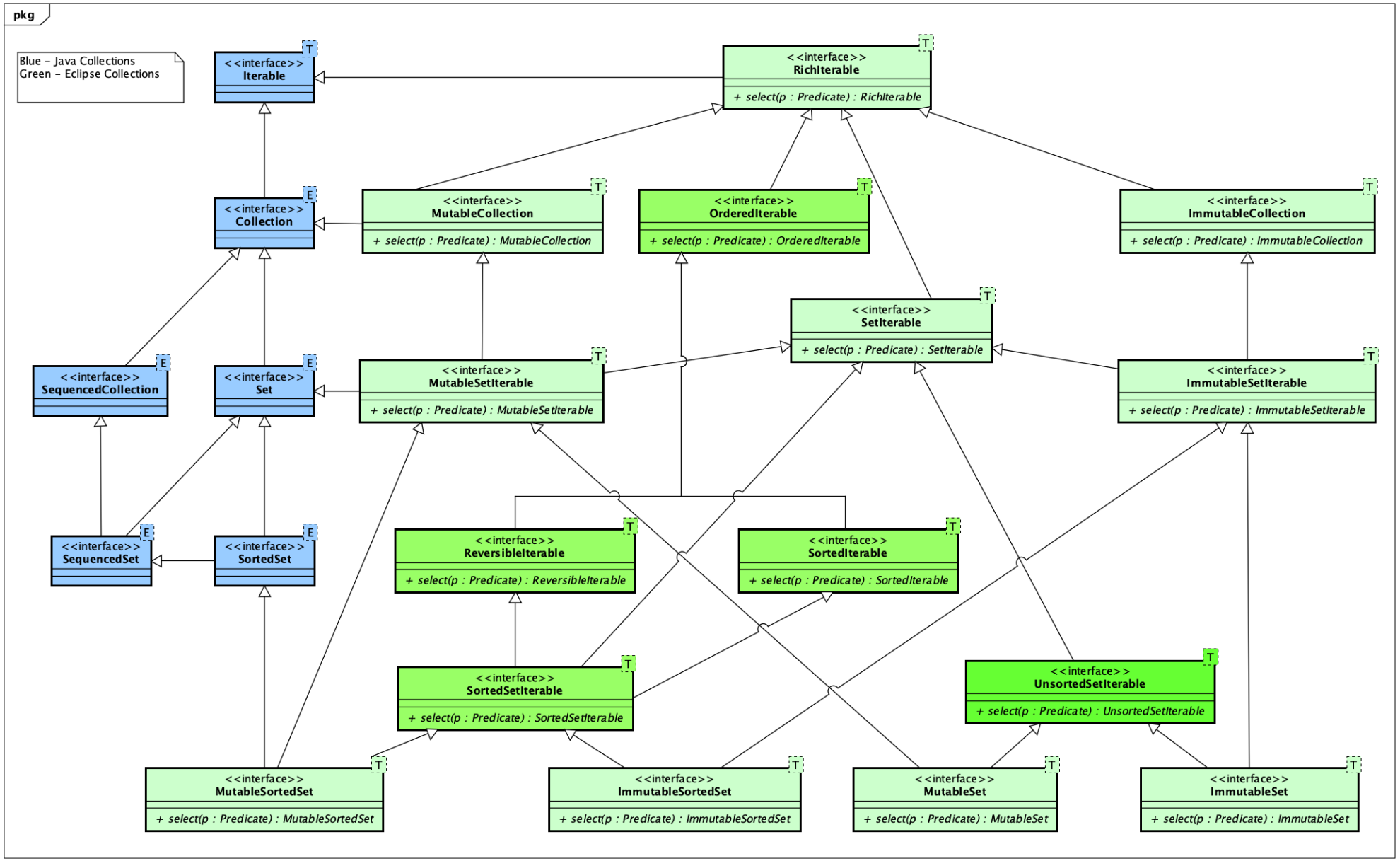
Takeaways

- Expect more from your collections
- Memory is an expensive resource – use it wisely
- Eclipse Collections – more features with less waste
- Intention revealing code – easy to write, easy to read!
- Many resources available
  - Source code w/ Reference Guide in AsciiDoc
    - https://github.com/eclipse-collections/eclipse-collections
  - Code katas
    - https://github.com/eclipse-collections/eclipse-collections-kata
  - Book: Eclipse Collections Categorically



Appendix

Eclipse Collections Set Type Hierarchy



Non-JDK Collection Types in EC